

KEVIN BUCK

GitHub: <https://github.com/kevmbuck>

Scholar: <https://scholar.google.com/citations?user=odO84I0AAAAJ&hl=en>

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EDUCATION

Indiana University, Bloomington, IN

Aug 2020 – Present

Ph. D. in Mathematics (*Expected*)

Advisors Roger Temam and Kevin Zumbrun

Wake Forest University, Winston-Salem, NC

Sept 2016 – May 2020

B.S. in Mathematics with Honors and Magna Cum Laude

Advisor John Gemmer

B.A. in Philosophy

PUBLICATIONS AND PREPRINTS

1. Kevin Buck and Roger Temam. “Convergence Properties of PINNs for the Navier-Stokes-Cahn-Hilliard System”. In: *Pure and Applied Functional Analysis* (2025). To appear. arXiv: [2505.07964](https://arxiv.org/abs/2505.07964) [[math.NA](#)]. url: <https://arxiv.org/abs/2505.07964>.
2. Kevin Buck and Woojeong Kim. “Auto-Adaptive PINNs with Applications to Phase Separation.” In: *Numerical Mathematics: Theory, Methods, and Applications* (2026). To appear. arXiv: [2510.23999](https://arxiv.org/abs/2510.23999) [[math.NA](#)]. url: <https://arxiv.org/abs/2510.23999>.
3. Kevin Buck and Kevin Zumbrun et al. *Nonconvex optimization and convergence of stochastic gradient descent, and solution of asynchronous game*. 2025. arXiv: [2503.02155](https://arxiv.org/abs/2503.02155) [[Math.NA](#)]. url: <https://arxiv.org/abs/2503.02155>.
4. Kevin Buck and Kevin Zumbrun et al. *Synchronous vs. Asynchronous Coalitions in Multiplayer Games, with Applications to Guts Poker*. 2024. arXiv: [2412.19855](https://arxiv.org/abs/2412.19855) [[cs.GT](#)]. url: <https://arxiv.org/abs/2412.19855>.
5. Kevin Buck and Kevin Zumbrun et al. *Continuous Guts Poker and Numerical Optimization of Generalized Recursive Games*. 2022. arXiv: [2208.02788](https://arxiv.org/abs/2208.02788) [[math.OC](#)]. url: <https://arxiv.org/abs/2208.02788>.

SEMINARS AND TALKS

IU PDE and Applied Math Seminar

Fall 2025

Physics Informed Neural Networks for Two-Phase Fluids

IU Math Graduate Student Seminar

Spring 2025

Physics Informed Neural Networks for Two-Phase Fluids

Centro Internazionale Matematico Estivo (CIME), Attended

July 2024

Diffuse Interface methods in Continuum Mechanics: analysis, singular limits and algorithms

IU PDE and Applied Math Seminar

Spring 2023

Introduction to Neural Networks for PDE Simulation

Institute for Scientific Computing and Applied Math Seminar (Series)

Spring 2023

Nonconvex Optimization and Physics Informed Neural Networks

IU Math Graduate Student Seminar

Fall 2021

Analysis of Physics Informed Neural Networks

TEACHING AND ACADEMIC WORK EXPERIENCE

Indiana University Bloomington, Bloomington, IN

Instructor of Record

- E201 – Linear Algebra for Data Science
 - Additionally developed a new curriculum for the course.
- X201 – Transition to Calculus

Spring 2025

Fall 2022, Spring 2024

- Additionally developed a new curriculum for the course.
- J111 – Introduction to College Math Fall 2021, Fall 2023 (2 Sections)
- M106 – Mathematics of Decision and Beauty Spring 2022, 2023

Lecture Series

- M741 – Topics in Applied Mathematics Spring 2024
 - Gave a series of lectures covering nonconvex optimization methods and machine learning.

ISCAM Server Administrator

- Managed the Institute for Scientific Computing and Applied Math's high performance Linux machine.
- Responsibilities included selecting the computers components, managing batch jobs and queues, keeping packages and other software updated, and managing accounts and security for the system with assistance from other IU organizations.

Log(IU)/LEMMA Coordinator

Spring 2022 – Fall 2025

- Led Log(IU), a program where graduate students mentor small groups of undergraduates on a specialized research project. Now renamed to LEMMA.
- Personally mentored a dozen students on many projects ranging from modeling herd dynamics of cows to analyzing of population dynamics of societies during war over five years as a graduate mentor.

REU Mentor

May – August 2023, 2024

- Assisted in mentoring two summer REU projects on nonconvex optimization and Game Theory with Kevin Zumbrun, leading to the preprints 4. and 5.

AWARDS AND OTHER GRANTS

Research Assistantship

June 2021 – Current

- Research Assistantship under Kevin Zumbrun
 - Received funding for the Fall 2025 Semester as well as Summer 2024.
- Research Assistantship under Roger Temam
 - Received funding for the Fall 2024 Semester as well as Summers 2021-2024.
- Mentored two summer REU programs about Nonconvex Optimization and Game Theory in 2023 and 2024

David A. Rothrock Fellowship for Outstanding Teaching

2025

- Awarded by IU Math Department for excellence in teaching.

Graduate Student Service Award

2025

- Awarded by IU Math Department for service to the surrounding mathematical community.

Hazel King Thompson Fellowship

2022, 2025

- Awarded by IU Math Department to cover summer expenses.

ISC Travel Award

2024

- Received grant to cover all travel expenses to travel for the CIME School in Cetraro, Italy.

CIME Grant

2024

- Received grant to cover all living expenses and board for the CIME School in Cetraro, Italy.

College Graduate Fellowship

2020-2021

- Awarded by Indiana University to cover graduate stipend for the first year of my Ph. D.

OTHER WORK EXPERIENCE

Price Waterhouse Coopers

New York, NY

Actuarial Intern

June 2019 – Aug 2019

- Involved in client work regarding data analysis of pensions, health insurance, and retirement plans.
- Adapted quickly to client needs in a fast-paced, detail-oriented environment.

Willis Towers Watson

Philadelphia, PA

Actuarial Intern

June 2018 – Aug 2018

- Used SQL to analyze several million rows of quantitative client data for initiative research.